Problem

The circuit shown in Fig. 14.98 has the impedance

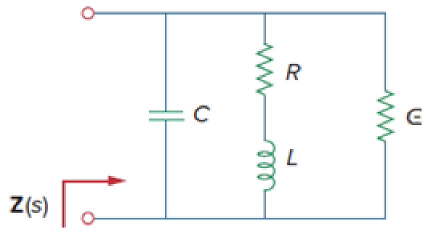
$$Z(s) = \frac{1{,}000(s+1)}{(s+1+j50)(s+1-j50)}, \qquad s = j\omega$$

Find:

(a) the values of *R*, *L*, *C*, and *G*

(b) the element values that will raise the resonant frequency by a factor of 103 by frequency scaling

Figure 14.98:



Step-by-step solution

Step 1 of 2

(a) $\frac{1}{Z} = G + j\omega C + \frac{1}{R + j\omega L} = \frac{(G + j\omega C)(R + j\omega L) + 1}{R + j\omega L}$

which leads to Z= $\frac{j\omega L + R}{-\omega^2 LC + j\omega(RC + LG) + GR + 1}$

$$Z(\omega) = \frac{j\frac{\omega}{C} + \frac{R}{LC}}{-\omega^2 + j\omega\left(\frac{R}{L} + \frac{G}{C}\right) + \frac{GR+1}{LC}} \qquad -(1)$$

We compare this with the given impedance $Z(\omega) = \frac{1000(j\omega + 1)}{-\omega^2 + 2j\omega + 1 + 2500} \quad -(2)$

comparing $(1)\,and\,(2)$ shows that

$\frac{1}{C} = 1000 \rightarrow C = 1MF, \quad \frac{R}{L} = 1 \rightarrow R = L$

$\frac{R}{L} = \frac{G}{C} = 2 \rightarrow G = C = 1ms$

$2501 = \frac{GR+1}{LC} = \frac{10^{-3}R + 1}{10^{-3}R} \rightarrow R = L = 0.4$

$thus, R = 0.4\Omega, \quad L = 0.4H_z, \quad C = 1MF, \quad G = 1ms$

Step 2 of 2

(b) By frequency - scaling, $K_f = 1000$

$R' = 0.4\Omega, \quad G = 1ms$

$$L' = \frac{L}{K_f} = \frac{0.4}{10^3} = 0.4MH_z$$

$$C' = \frac{C}{K_f} = \frac{10^{-3}}{10^{-3}} = 1\mu F$$